

Question ID: f76c1858

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$7x^2 - 20x - 32 = 0$

What is the positive solution to the given equation?

Correct Answer: 4

Rationale

The correct answer is 4. The left-hand side of the given equation can be factored as $(7x + 8)(x - 4)$. Therefore, the given equation, $7x^2 - 20x - 32 = 0$, can be written as $(7x + 8)(x - 4) = 0$. Applying the zero product property to this equation yields $7x + 8 = 0$ and $x - 4 = 0$. Subtracting 8 from both sides of the equation $7x + 8 = 0$ yields $7x = -8$. Dividing both sides of this equation by 7 yields $x = -\frac{8}{7}$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the two solutions to the given equation, $7x^2 - 20x - 32 = 0$, are $-\frac{8}{7}$ and 4. It follows that 4 is the positive solution to the given equation.